

Daniel Santiago Fleming

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EDUCATION

Massachusetts Institute of Technology | *BS in Computer Science and Engineering*

Organizations: Society of Hispanic Professional Engineers, TAMID Group, Chess Club, Poker Club

SKILLS

Programming Languages: Python, TypeScript, C++, R

Machine Learning: PyTorch, vLLM, TensorFlow, OpenCV

Software / Frameworks: Next.js, Tailwind CSS, React, FastAPI, Flask, LangGraph, Pipecat

Tools / Tech: PostgreSQL, Supabase, Prisma, Git, Bash, GCP, AWS, Docker, CI/CD

Languages: Spanish (ILR Level 4)

TECHNICAL EXPERIENCE

Netic AI | *Founding Engineer*

November 2024 - Present

- #2 Engineering Hire at Founders Fund + Greylock backed AI Startup
- Built from scratch the company's AI Agent, driving ARR from \$250K → \$10M in under 12 months.
- Scaled the Agent to autonomously handle 4 million+ interactions, generating over \$500M in revenue for customers
- Owned the full agent stack end to end: orchestration graph, voice pipeline, tool integrations, and evaluation harness

LLM Inference Engine | *PyTorch, Modal*

July 2026 - August 2026

- Built an LLM inference engine from scratch, reaching 51% of vLLM's throughput serving Qwen2.5-1.5B on one A10 GPU
- Implemented continuous batching, paged attention, chunked prefill, and a FlexAttention decode kernel
- Cut decode stalls from 2,874 ms → 78 ms, measuring each optimization in isolation
- Served engine on Modal, scaling throughput 20x from 1-32 concurrent requests, publishing latency and throughput results

Personalized Chess Engine | *Python, Tensorflow, Google Cloud Platform, Flask*

Aug 2024 - Sep 2024

- Trained a CNN policy network on 1M+ Lichess games filtered to my rating range to predict human moves
- Fine-tuned on my own games to reproduce my opening repertoire, training on an NVIDIA L4 and serving over Flask

Quarterly Report Parser | *TypeScript, Next.js, Supabase*

October 2024

- Developed a full-stack web application using Next.js for user authentication and Supabase PostgreSQL for storage
- Parsed uploaded 10-Q reports utilizing LLM-based data parsing to automate financial data extraction into CSV format

Enhancing Stable Diffusion with ControlNet | *Python, OpenCV*

Apr 2024 - May 2024

- Fine-tuned Stable Diffusion by integrating ControlNet to incorporate color information by conditioning the model on colored edge maps alongside text prompts to improve image fidelity
- Improved image quality by 11% using Color Mean Squared Error (MSE) to compare an enhanced model against a baseline Stable Diffusion model

AI Mammogram Analysis | *Python, OpenCV, PyTorch*

Mar 2024 - Apr 2024

- Improved breast cancer detection to 93.4% validation accuracy by training a Mask R-CNN model for classifying and segmenting mammogram images
- Evaluated model performance, achieving recall of 1.0 with confusion matrices and precision-recall curves

Computational Analysis of Hi-C and RNA-seq Datasets | *Python, NumPy, Sklearn, Pandas*

Mar 2024

- Identified interchromosomal interactions in Hi-C data by detecting 442 high-interaction regions ($p < 0.01$) by developing a greedy search algorithm
- Identified key genomic markers for cell clusters using PCA and regularized logistic regression on single-cell RNA-seq data

RESEARCH EXPERIENCE

MIT Sports Lab | *Undergraduate Researcher*

Sep 2023 - May 2024

- Enhanced NBA injury risk prediction by a 14% accuracy improvement by analyzing NBA skeletal tracking data

MIT Digital Humanities Lab | *Undergraduate Researcher*

June 2021 - July 2022

- Accelerated the formatting of over 100,000 images by 50% through implementing a multithreaded architecture

Summer Science Program (SSP) Astrophysics | *Researcher*

June 2019 - July 2019

- Determined the orbital elements of asteroid (2001 XT30) with 99.6% accuracy via Gauss's method, Monte Carlo analysis, and Differential Correction; Published asteroid data to the Minor Planet Center by submitting verified observations

PUBLICATIONS

New Observations of Near-Earth Asteroid 153842 (2001 XT30)

June 2019 - July 2019

- Used astrometry and photometry to analyze data collected from the asteroid before calculating its trajectory. Analysis of its orbit and information from JPL Horizons shows that it's a Mars-crossing asteroid with unlikely potential to hit Earth soon
- Minor Planet Center, July 27, 2019; Rolfness, J., Liang, L., Fleming, D., et al. 2019 "New Observations of Near-Earth Asteroid 153842 (2001 XT30)," M.P.S. 906049, M.P.C 11064